Michael Warren
4/28/2019
GEOG432: Advanced GIS
Spring 2019
Final Project
Instructor: Dr. Hua Liu

Windmills of Texas and Averaged Windspeeds of March 2019

Section I: Introduction

Development of renewable energy sources is an essential component to diversifying the energy profile of the United States. Rapidly growing industries such as wind and solar in Texas (Chang and Starcher, 2019) have helped drive out societies' reliance on conventional energy such as oil, coal, and natural gas. The scientific consensus on climate change (Cook et al., 2016) motivates to develop such resources as fossil fuel emissions continue to impact climate change (IPCC, 2014). However, where conventional energy sources are often referred to as *extractive*, only needing to be removed from their place of origin to be used, the availability of wind and solar energy varies with time; hence, there is a critical need to investigate wind current data over time to generate better predictions and thus higher reliability in energy production.

Texas is the leader in wind energy development and power generation in the United States (Chang and Starcher, 2019). With abundant wind energy resources, wind currents in Texas remains an essential area of study.

Windmills need a constant supply of wind energy in order to maximize energy output and avoid stalling the rotors. Wind energy can be defined and determined by the following equation:

$$P = \frac{1}{2}\rho v^3 \pi r^2$$

Source: Danish Wind Industry Association (2012)

Where v is the wind speed and the other variables are constants; thus, a doubling of the speed of the wind, generates eight times more power (Danish Wind Industry Association, 2012). The objective of this study was to indicate if ground measurements of wind speed data correlated with the spatial positioning of windmills in Texas.

Section II: Methodology

The study area for this project was the entire state of Texas. This area was selected given its vast energy resources, its spatial extent, and availability of data. The month of March was selected as it was the most recent month to date.

Iowa State University provides an excellent portal for Automated Surface Observing System (ASOS) weather METAR data and thus was the basis for extracting wind speeds. METAR data includes wind velocities at, usually, 5-minute intervals. ASOS METAR data was downloaded between the dates of February 26, 2019, and April 2, 2019. The margins of the data were then excluded, within Excel, in order to verify all measurements within March were gathered.

Figure 1: Excluded station VKY Zapata-Yorktown due to out of range

The entire dataset downloaded from the portal included 226 stations. Seventeen stations were excluded from the dataset; 16 were excluded because they reported ("None") under the attribute of ("avg_wind_speed_kts") and 1 was excluded, named ("VKY Zapata-Yorktown") seen above in Figure 1, because it clearly fell outside of the spatial extent of Texas.

The resulting stations' monthly average wind speed was computed 3/1/2019 (UTC) through 3/31/2019 (UTC) for each station, respectively.

The tool that I used for this study was the Inverse Distance Weighting for the interpolation. IDW interpolation assumes that spatial correlation decreases with distance, and thus weights its approximation inversely with distance. This is appropriate for wind data, as the wind speed at location A may give a hint as to the wind speed at nearby location B, but is not very useful for location C. The problem is that wind is so spatially variable that it's hard to determine what influences it may have for location C.

Another tool I utilized for this project was the 3D representation in ArcScene to demonstrate the topography of Texas.

Colorbrewer2 (Brewer, 2019) was used for the color ramp generations.

Section III: Results

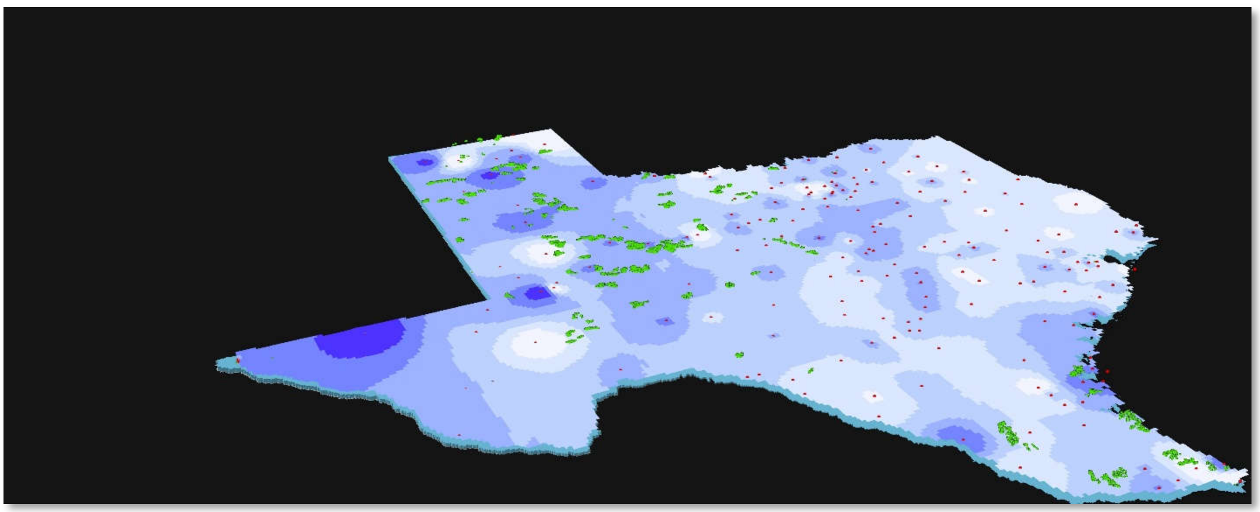
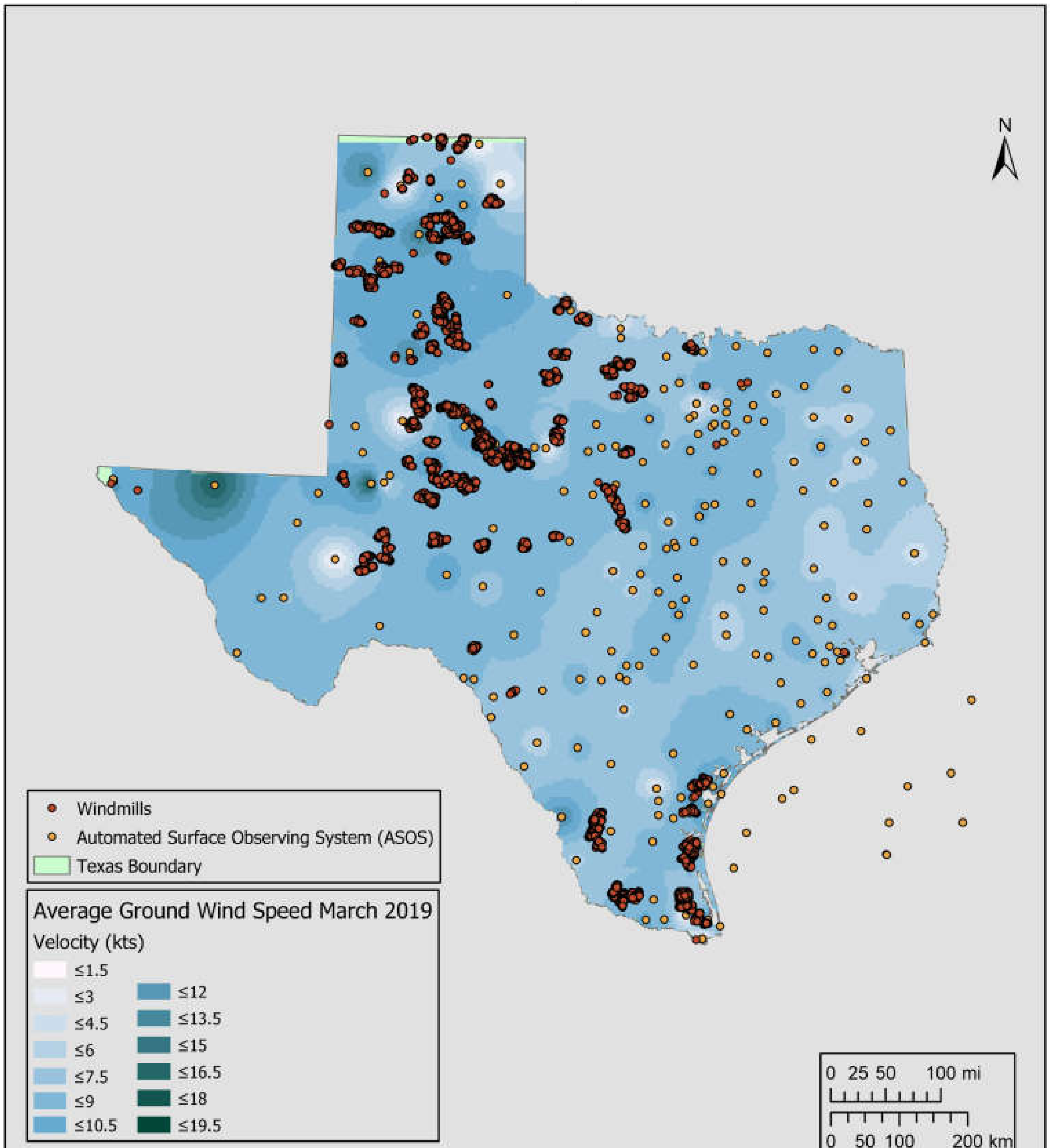


Figure 2: ArcScene representation of the interpolated ground windspeeds of Texas for March 2019

The above image is a 3D representation of Texas overlaid on a 30-meter DEM. I have intentionally left the legend out of the image to keep the artistic integrity of the image. This image is not for analysis or interpretation, and only for artistic ability.

Texas Ground Wind Speeds March 2019



Source: Iowa State University, University of Texas at Austin, USGS, TxDOT

Spatial Reference:
PCS: NAD 1983 Texas Statewide Mapping System
GCS: NAD 1983 Texas Statewide Mapping System
Datum: North American 1983
Projection: Lambert Conformal Conic

Michael Warren
4/28/2019

Section IV: Discussions and Conclusion

It can be seen in our analysis that wind speed data over large areas is available to a high degree of precision in both space and time, and can be analyzed and displayed using ArcGIS software and spatial analysis interpolation techniques. Generating a sound model in order to observe changes over time is essential to understanding spatial phenomena, and energy on wind data has a unique characteristic in that utilizing it will change its distribution. Large-scale wind energy collection via turbines will ‘drain’ the wind energy from one area and make changes in the distribution generated in Figure 2 and Figure 3; planning and research activities must be in a position to anticipate these changes to a high degree of spatial and numerical accuracy in order to assure efficient utilization of this precious resource.

While it was unfortunate that further analysis could not be done on the wind raster due to user limitations, it is vital to understand the power and ability to join large amounts of data, spatially join it, and effectively communicate the data. This project was an example of those GIS techniques. Utilizing Excel, formatting techniques, database management skills, and other multidisciplinary knowledge, 5967 wind measurements, on different days, were generalized in developing the above maps. GIS requires the knowledge of a vast range of skills and the communication of the data is crucial to the puzzle piece of geography.

Section V: References

Work Cited

Brewer, C., Harrower, M. and Pennsylvania State University (2019). *ColorBrewer: Color Advice for Maps*. [online] Colorbrewer2.org. Available at: http://colorbrewer2.org [Accessed 29 Apr. 2019].

Ruppel, S. (2008). *Permian Basin GIS Projects and Databases*. [online] Beg.utexas.edu. Available at: http://www.beg.utexas.edu/resprog/permianbasin/gis.htm [Accessed 29 Apr. 2019].

USGS, Berkeley Lab and AWEA (2019). *Data & Web Services*. [online] Eerscmap.usgs.gov. Available at: https://eerscmap.usgs.gov/uswtdb/data/ [Accessed 29 Apr. 2019].

Texas Department of Transportation (2015). *Texas State Boundary*. [online] Arcgis.com. Available at: https://www.arcgis.com/home/item.html?id=8cc5fa4c5cdb498bb74be021f3e07d02 [Accessed 29 Apr. 2019].

IPCC, 2014: Climate Change 2014: Synthesis Report. Contribution of Working Groups I, II and III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change [Core Writing Team, R.K. Pachauri and L.A. Meyer (eds.)]. IPCC, Geneva, Switzerland, 151 pp.

Cook, J., Oreskes, N., Doran, P., Anderegg, W., Verheggen, B., Maibach, E., Carlton, J., Lewandowsky, S., Skuce, A., Green, S., Nuccitelli, D., Jacobs, P., Richardson, M., Winkler, B., Painting, R. and Rice, K. (2016). Consensus on consensus: a synthesis of consensus estimates on human-caused global warming. *Environmental Research Letters*, [online] 11(4), p.048002. Available at: https://iopscience.iop.org/article/10.1088/1748-9326/11/4/048002/pdf [Accessed 28 Apr. 2019].

Chang, B. and Starcher, K. (2019). Evaluation of wind and solar energy investments in Texas. *Renewable Energy*, [online] 132, pp.1348-1359. Available at: https://doi.org/10.1016/j.renene.2018.09.037 [Accessed 28 Apr. 2019].

Department of Agronomy (2019). *IEM Computed Daily Summary of Observations*. [online] Mesonet.agron.iastate.edu. Available at: http://www.mesonet.agron.iastate.edu/request/daily.phtml?network=TX_ASOS [Accessed 27 Apr. 2019].

Danish Wind Industry Association (2003). *The Power of the Wind: Cube of Wind Speed*. [online] Xn--drmstrre-64ad.dk. Available at: http://xn--drmstrre-64ad.dk/wp-content/wind/miller/windpower%20web/en/tour/wres/enrspeed.htm [Accessed 29 Apr. 2019].

Section VI: Acknowledgements
My parents for their love and support
Old Dominion University
Dr. Hua Liu – For her support through this semester and dedication to her students
Dr. Michael Allen – for his enthusiasm and passion for teaching